Purpose

This is a step-by-step procedure on distributing the Offline Root CA's certificate to all [INTERNAL-ORG] domain computers. This ensures **Windows** devices will automatically trust certificates from the Intermediate CA as well as receive auto-enrollment.

Scope

In-Scope

- Group Policy Object to distribute Root CA certificate and auto enroll Windows devices

Out-of-Scope

- Manual install of certificates on non-domain joined or non-Windows devices

Requirements

Permissions: Domain Administrator

Access to Windows Domain Controllers and Online Intermediate CA

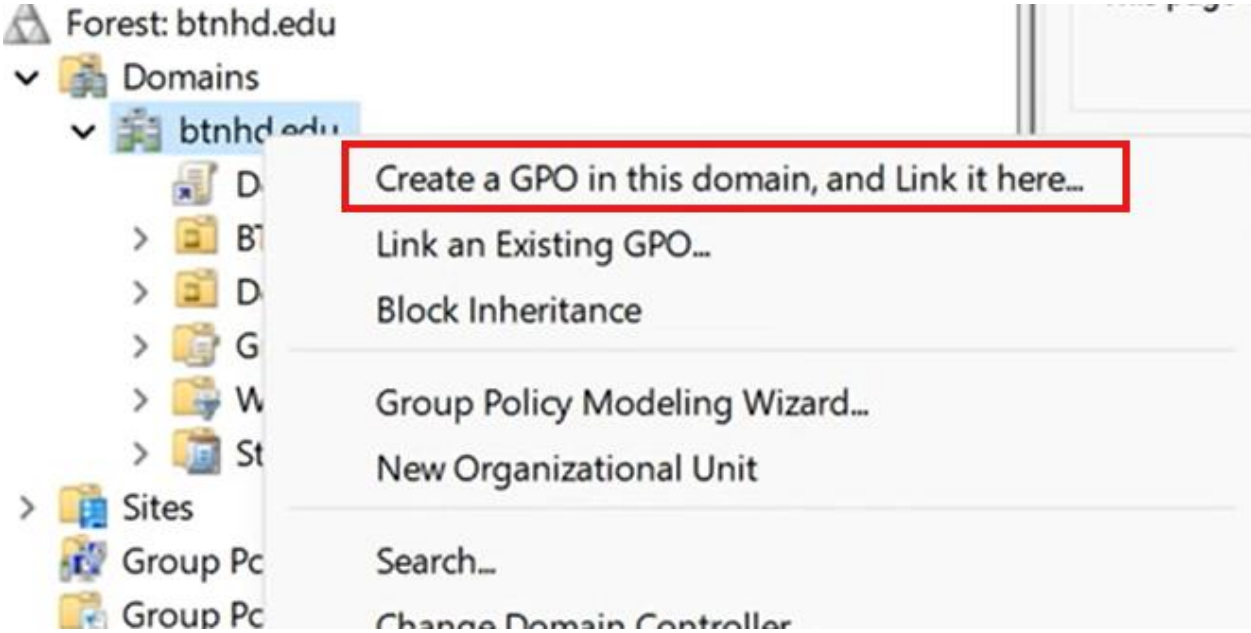
Ensure Domain workstations/servers can reach the Online Intermediate CA server

Procedure

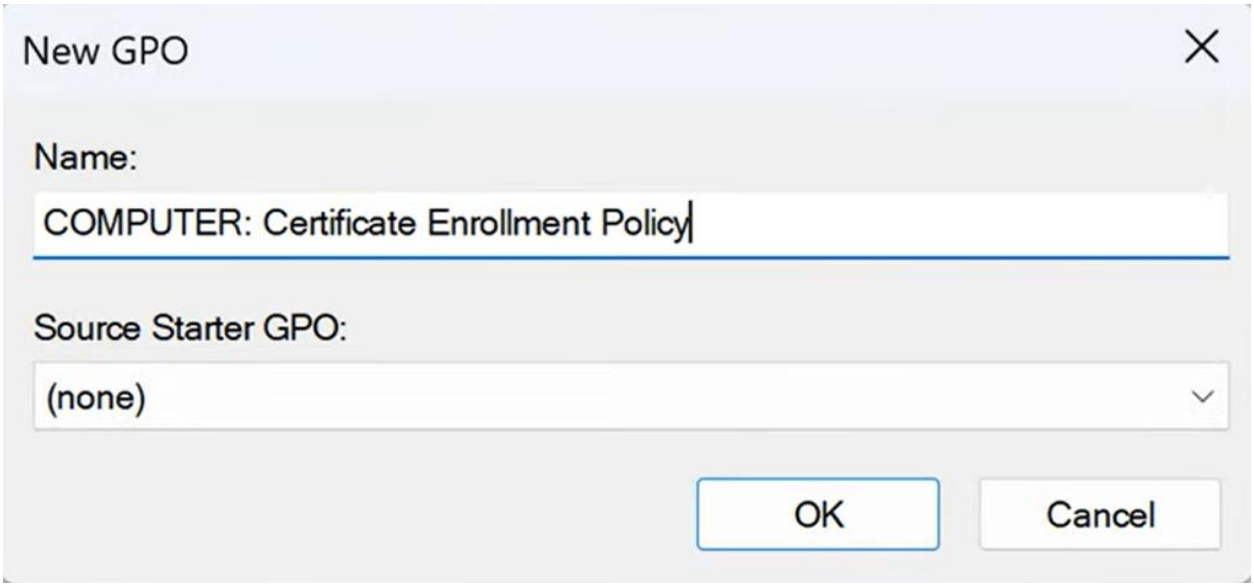
Creating the GPO in Domain Controller

1. Go to **Group Policy Management (gpmc.msc)**

2. **Right-click** on our **domain** and click **Create a GPO in this domain, and Link it here…**



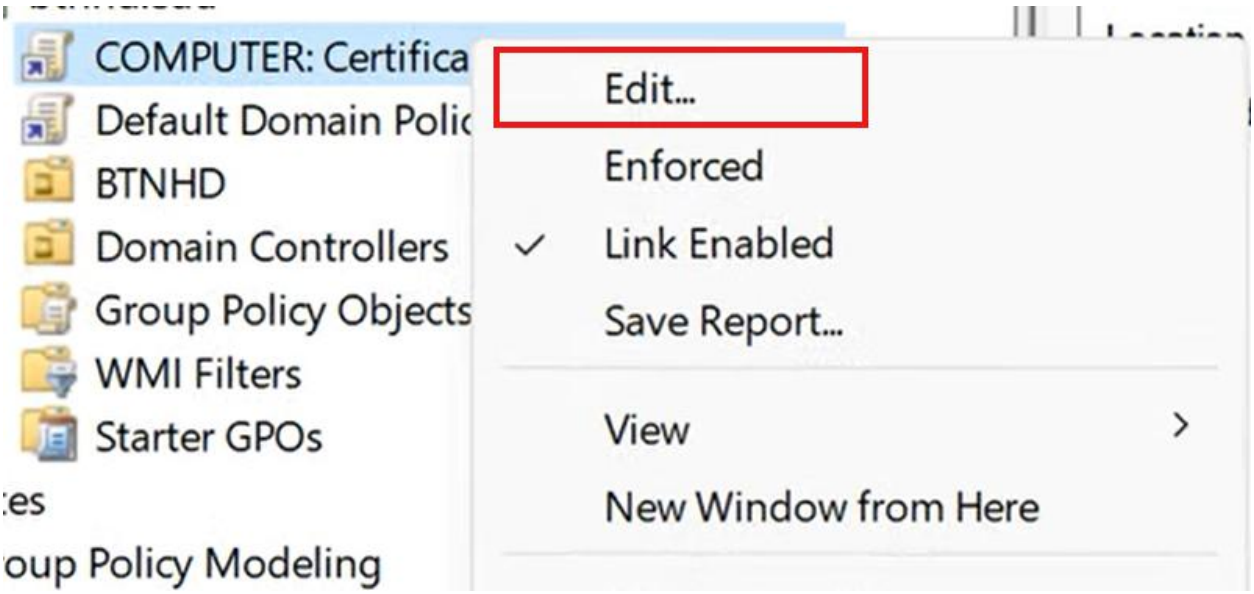
3. Set name as necessary, for clarity I recommend **[INTERNAL-ORG] PKI Distribution**



4. Click **OK**

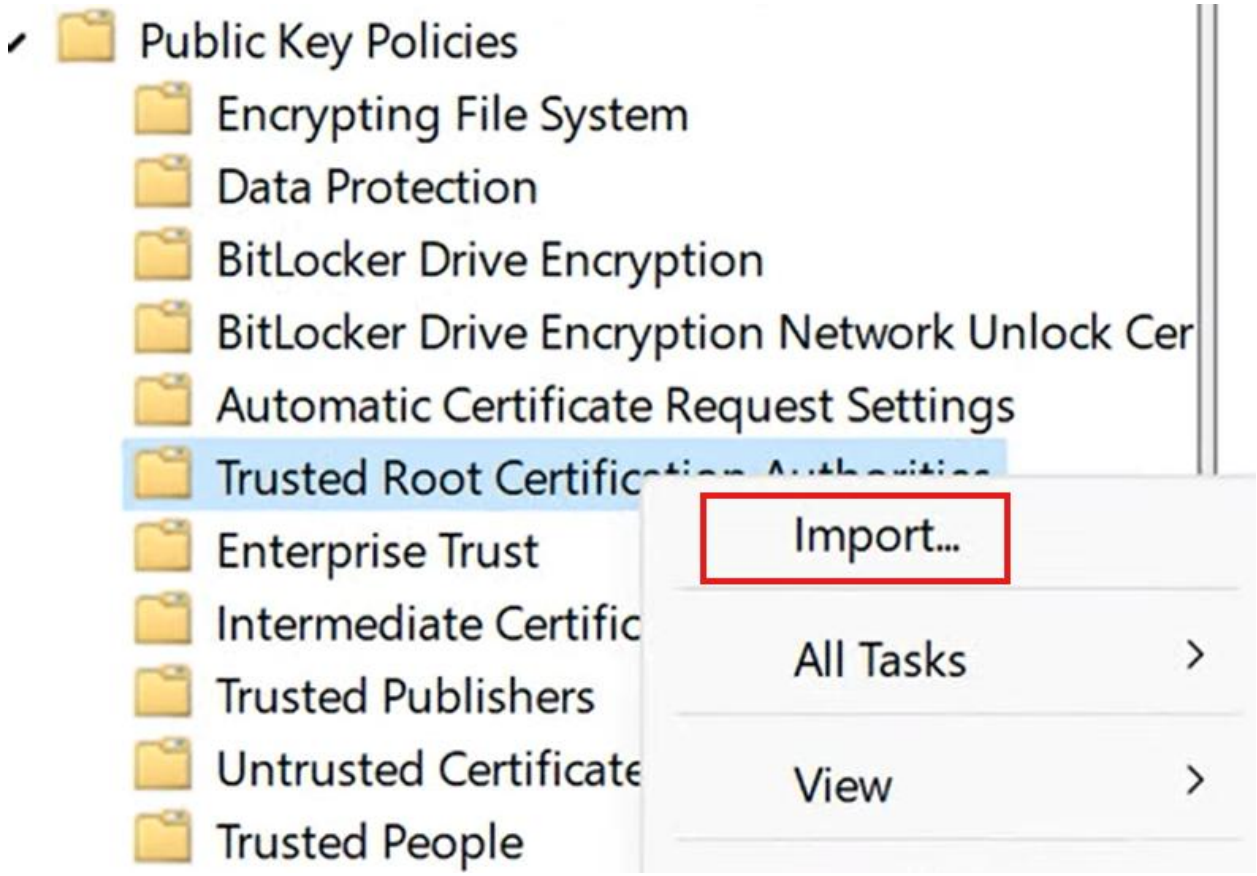
Importing the Root CA and Intermediate CA in the GPO

1. With the named GPO created **right-click** the GPO and click **Edit…**

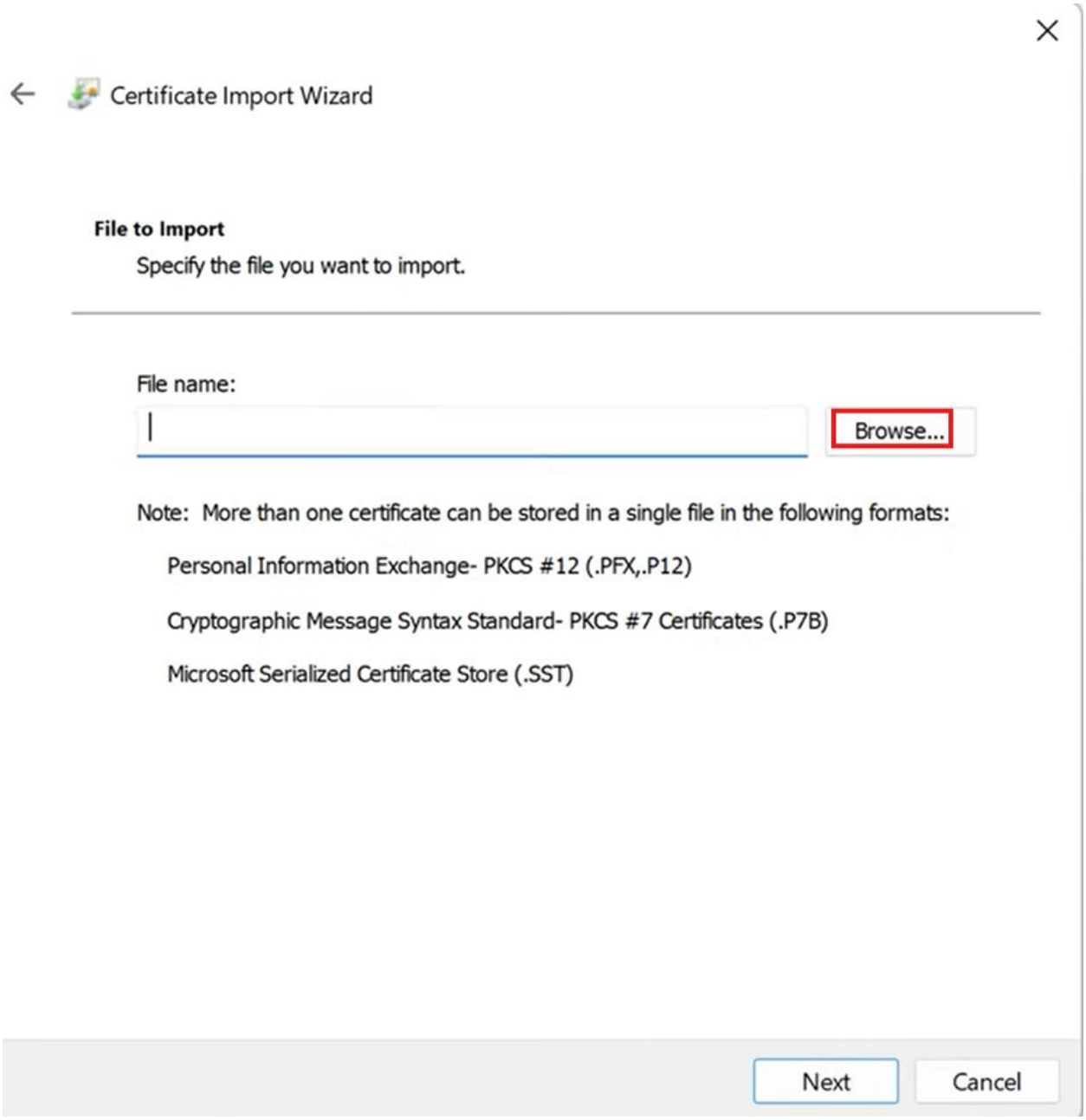


2. Within our GPO, go to **Computer Configuration > Policies > Windows Settings > Security Settings > Public Key Policies**

3. Right-click **Trusted Root Certification Authorities** and click **Import...**



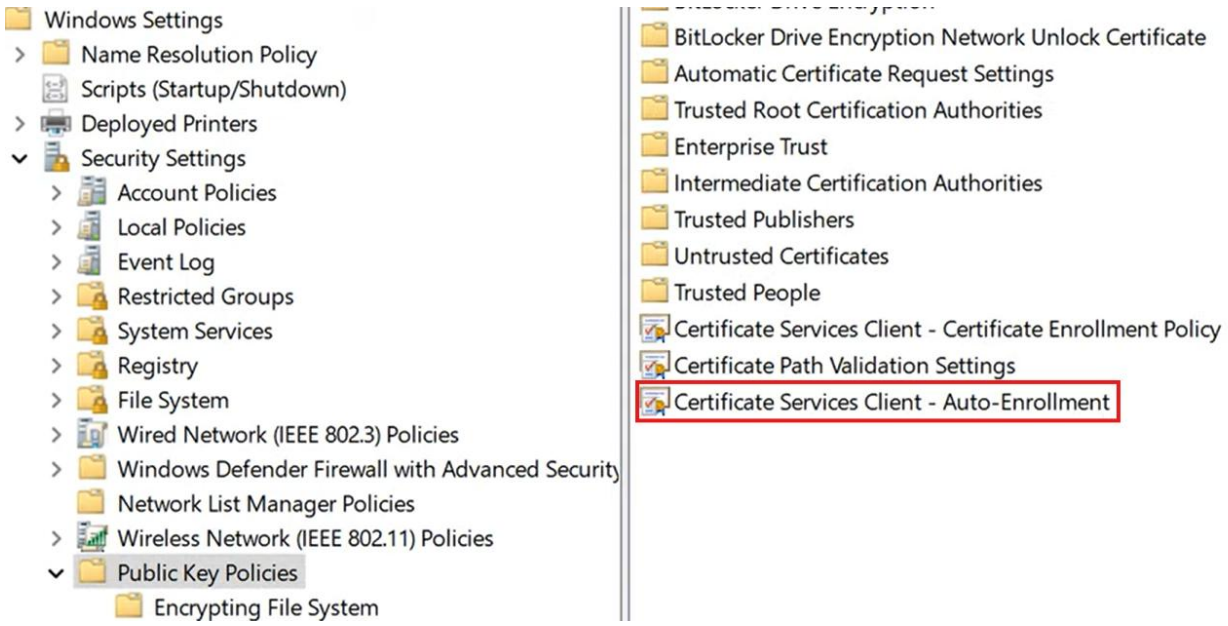
4. In the pop-up wizard, continue until you reach **Browse...**



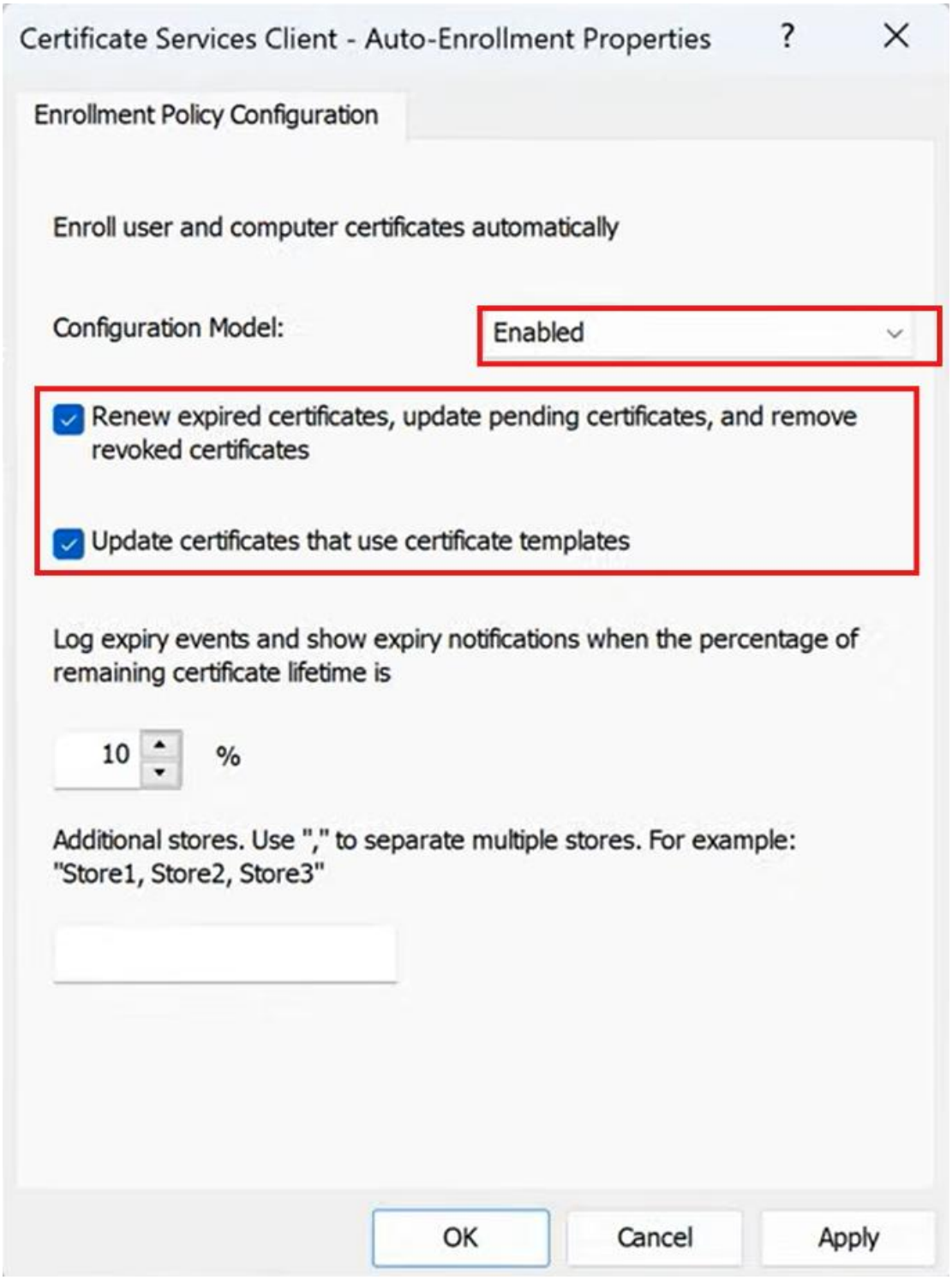
5. **Open** the **Root .cer** file that is located in the Intermediate CA server's **inetpub\wwwroot\CertEnroll** folder (may be done with secure media or network share)
6. Accept all wizard defaults and finish
7. Repeat steps 3-6 with the **Intermediate Certification Authorities** folder and importing the **Intermediate CA .cer file**

Enabling Auto-Enrollment Policy in the GPO

1. Within our named GPO, go to **Computer Configuration > Policies > Windows Settings > Security Settings > Public Key Policies**
2. Under Public Key Policies, double-click **Certificate Services Client – Auto-Enrollment**



3. **Enable** the policy and **Check** the boxes for **Renew expired…** and **Update certificates…**

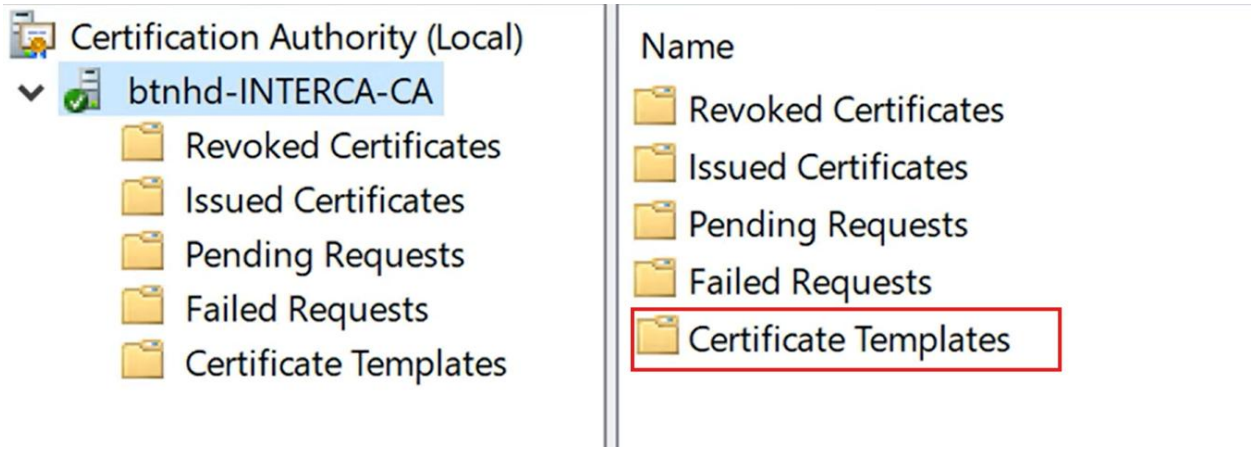


4. Click **Ok**

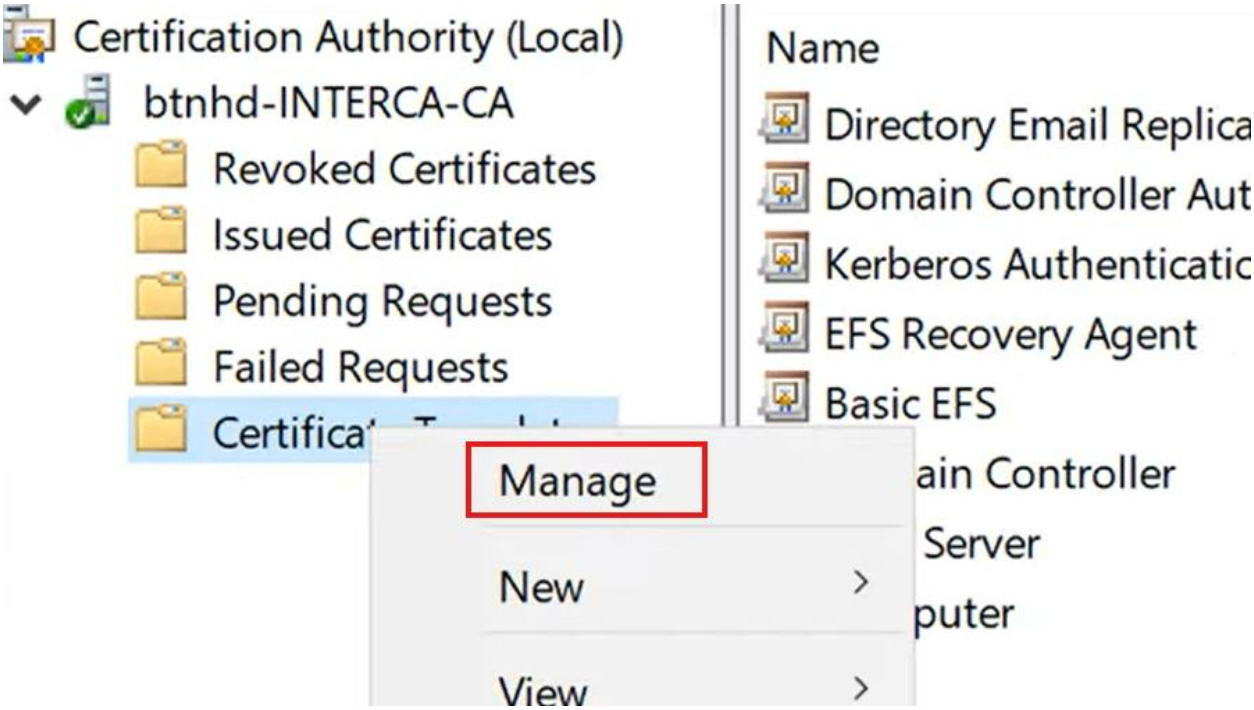
Creation of Windows Certificate Template

In the Intermediate CA server

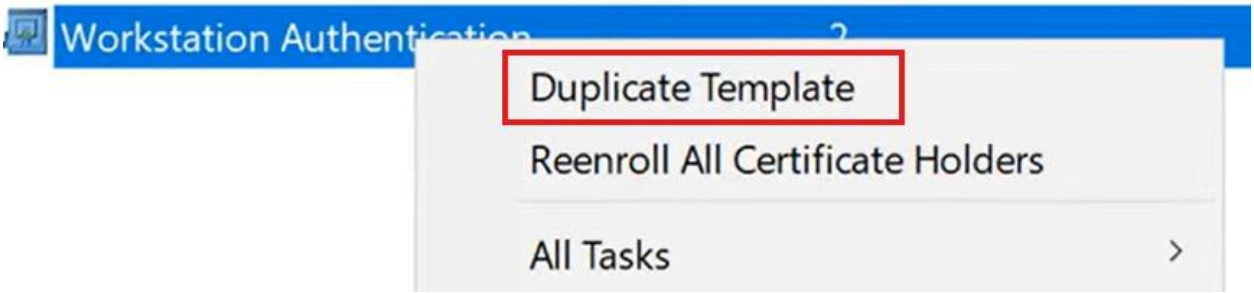
1. Open **Certification Authority (certsrv.msc)**
2. Under our server, go to **Certificate Templates**



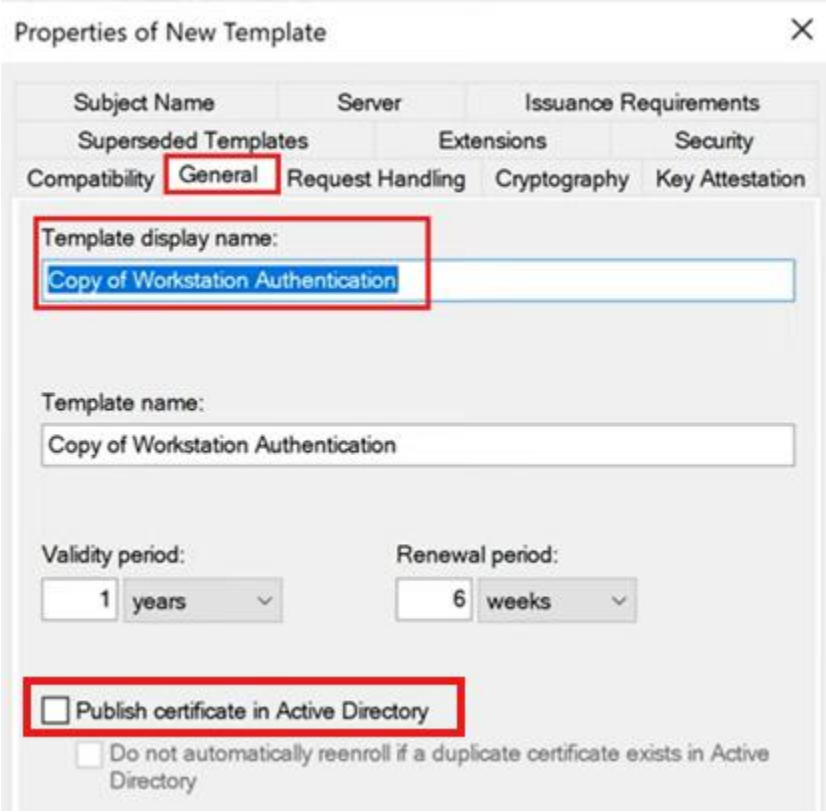
3. Right click **Certificate Templates** and click **Manage**



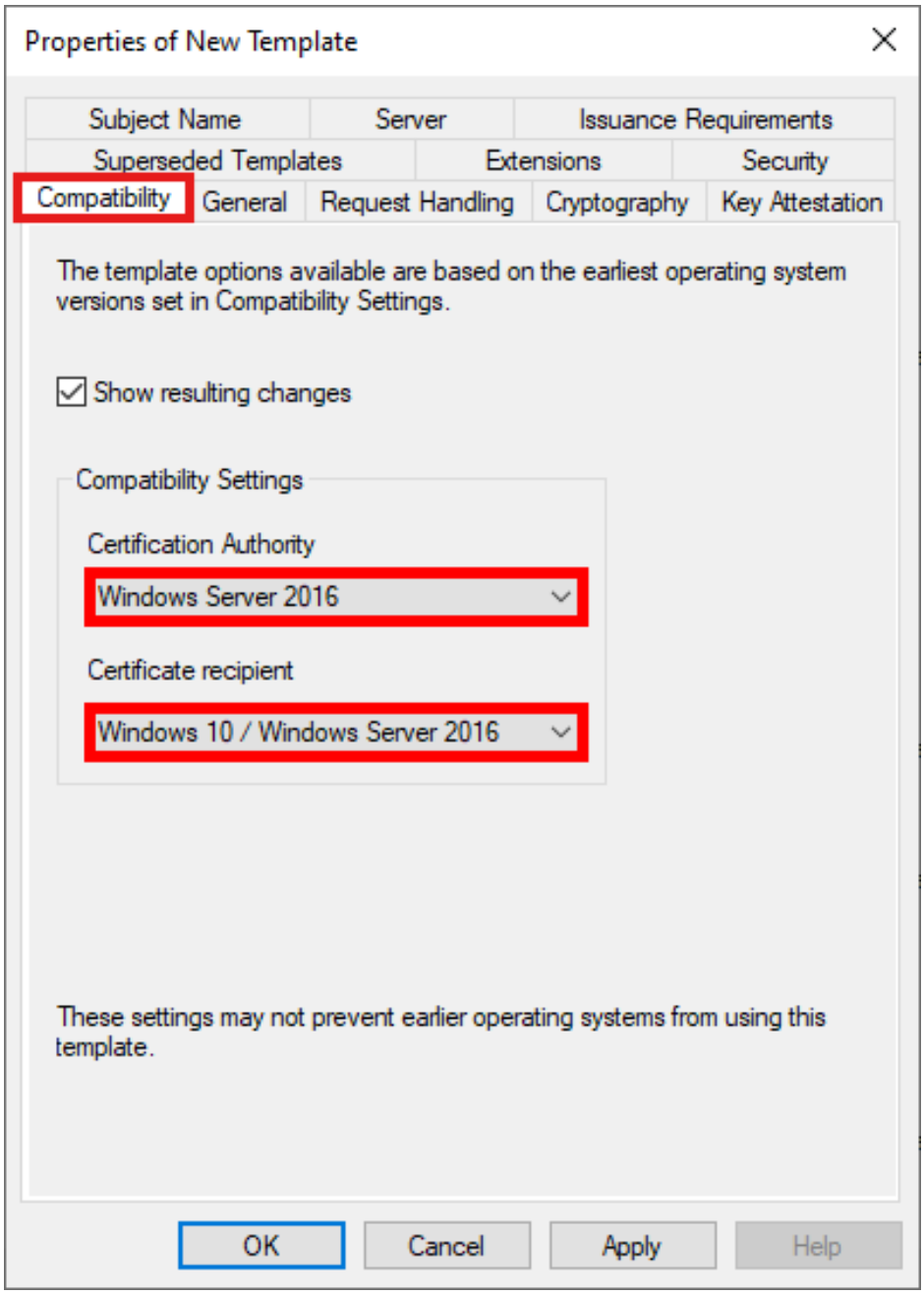
4. Right-click **Workstation Authentication** and click **Duplicate Template**



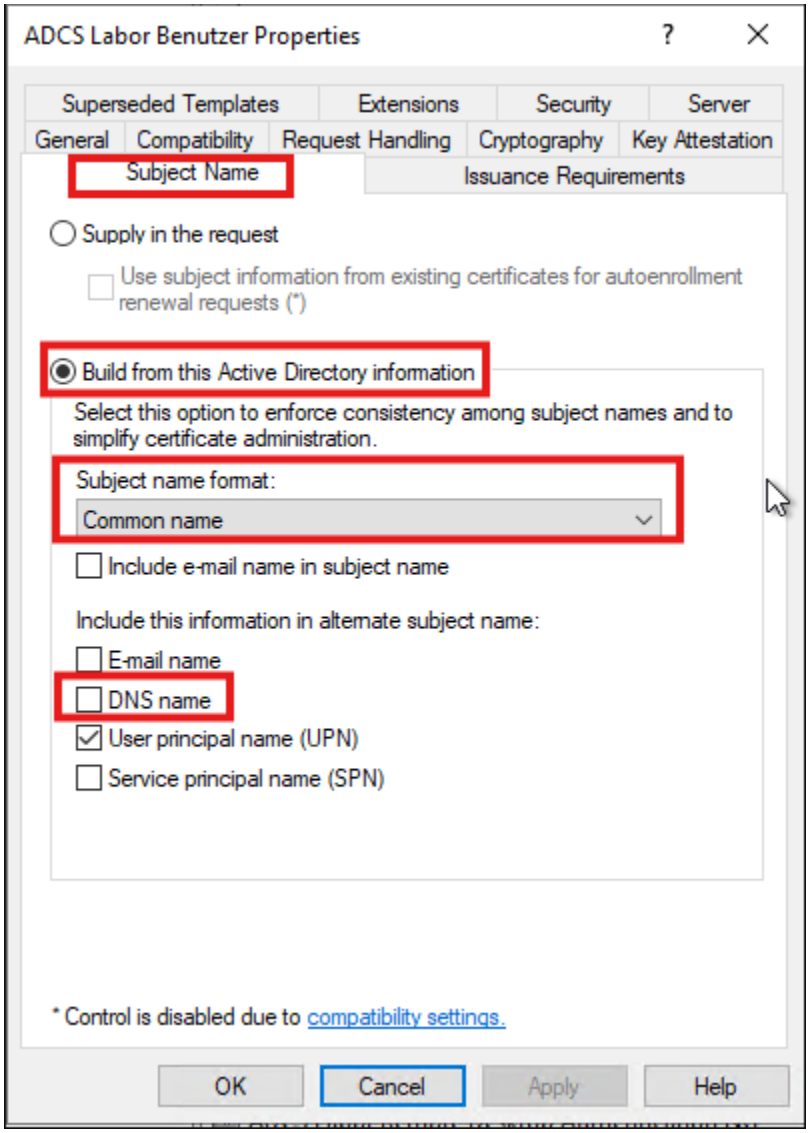
5. In the **General** tab,
   a. Name the template as necessary. I suggest **[INTERNAL-ORG] Windows Authentication**
   b. Set the **Validity Period** and **Renewal Period** as necessary for workstation/server certificates (preferably **2 year** and **6 weeks** respectively)
   c. Check **Publish certificate in Active Directory**



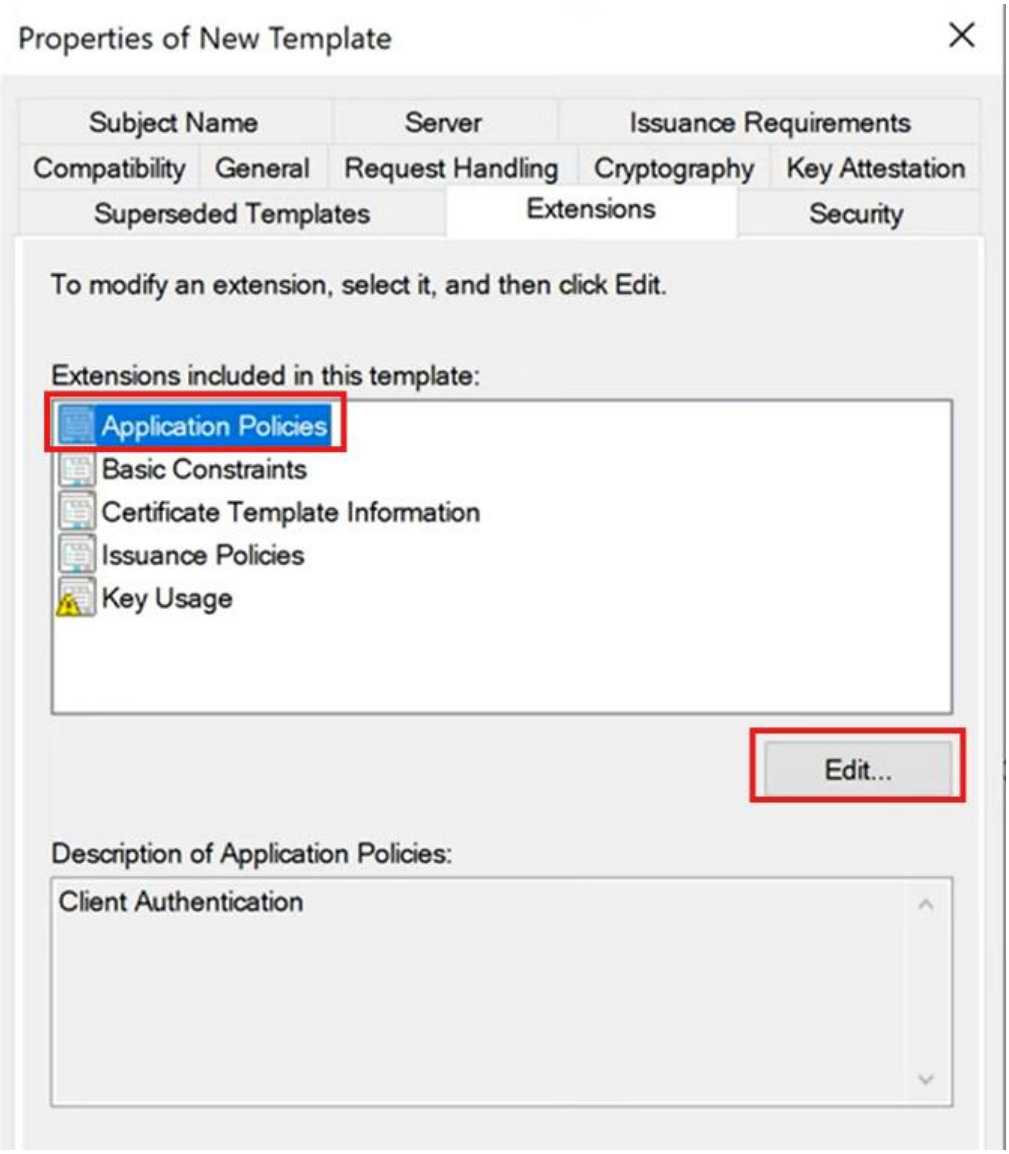
6. In the **Compatibility** tab,
    a. Set **Certification Authority** to **Windows Server 2008 R2** (must be compatible with oldest server)
    b. Set **Certificate Recipient** to **Windows 7 / Server 2008 R2** (must be compatible with oldest server)



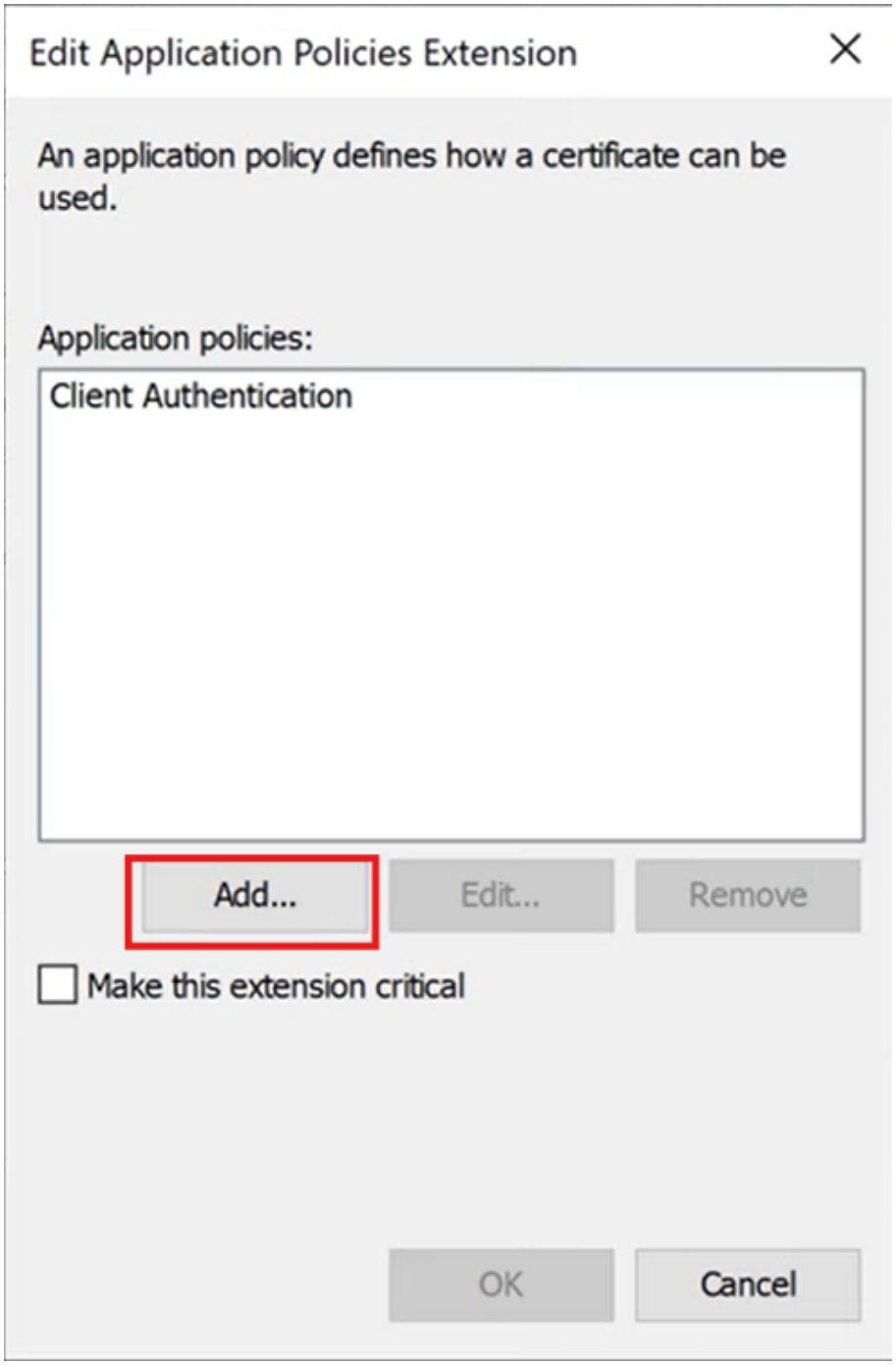
7. In the **Subject Name** tab,
    a. Select **Build from this Active Directory information**
    b. Change **Subject name format** to **Common name**
    c. Check the box for **DNS name**



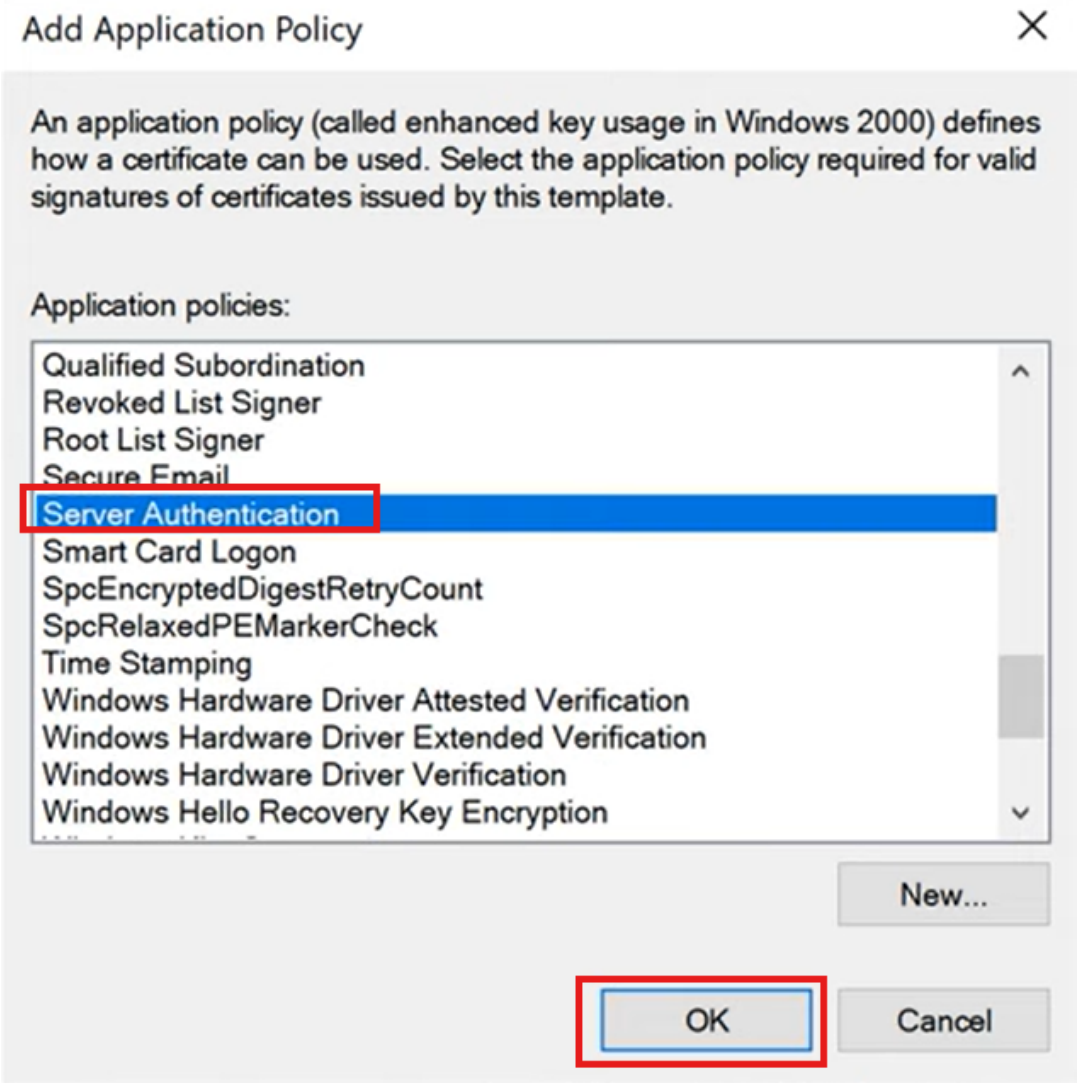
8. In the **Extensions** tab,
    a. With **Application Policies** selected, click **Edit**



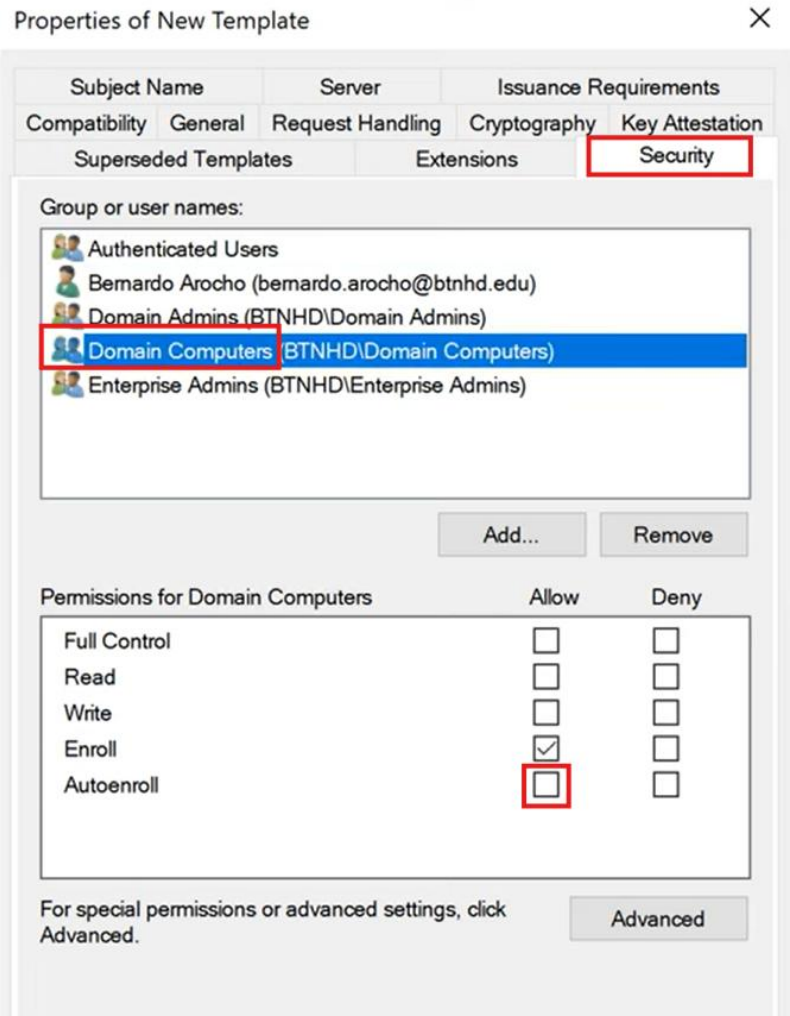
b. In the **Edit Application Policies Extension** window**,** click **Add**



c. In the **Add Application Policy** window, select **Server Authentication** and click **Ok**



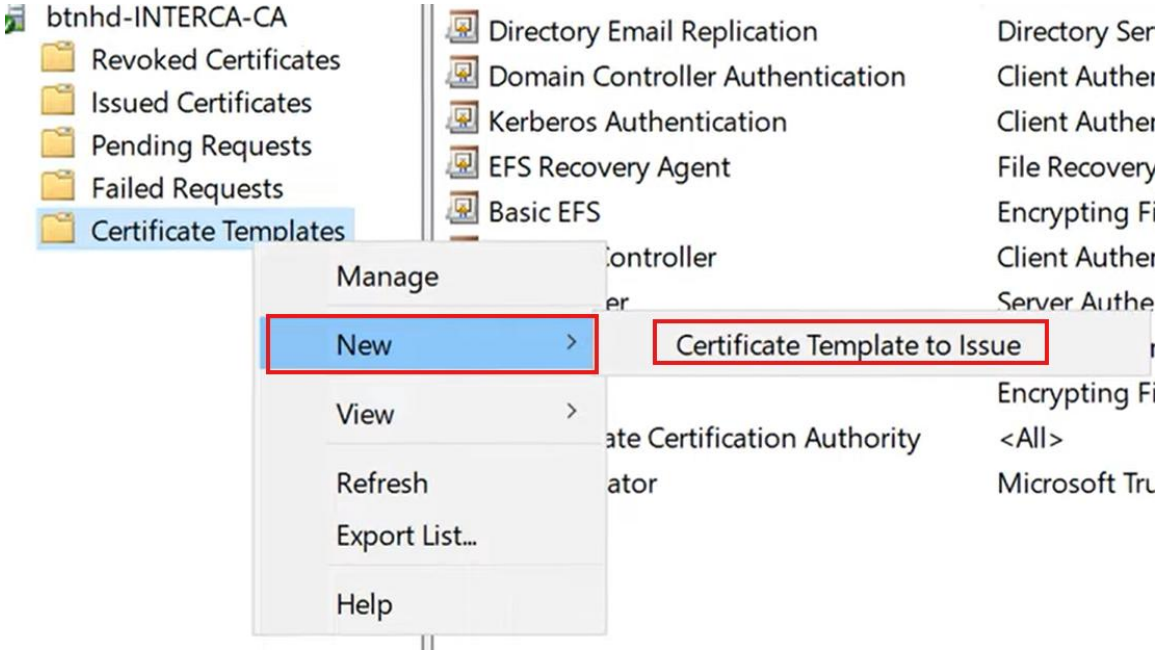
9. In the **Security** tab,
    a. For the group **Domain Computers**, ensure the permissions, **Read**, **Enroll**, and **Autoenroll** are checked to **Allow**



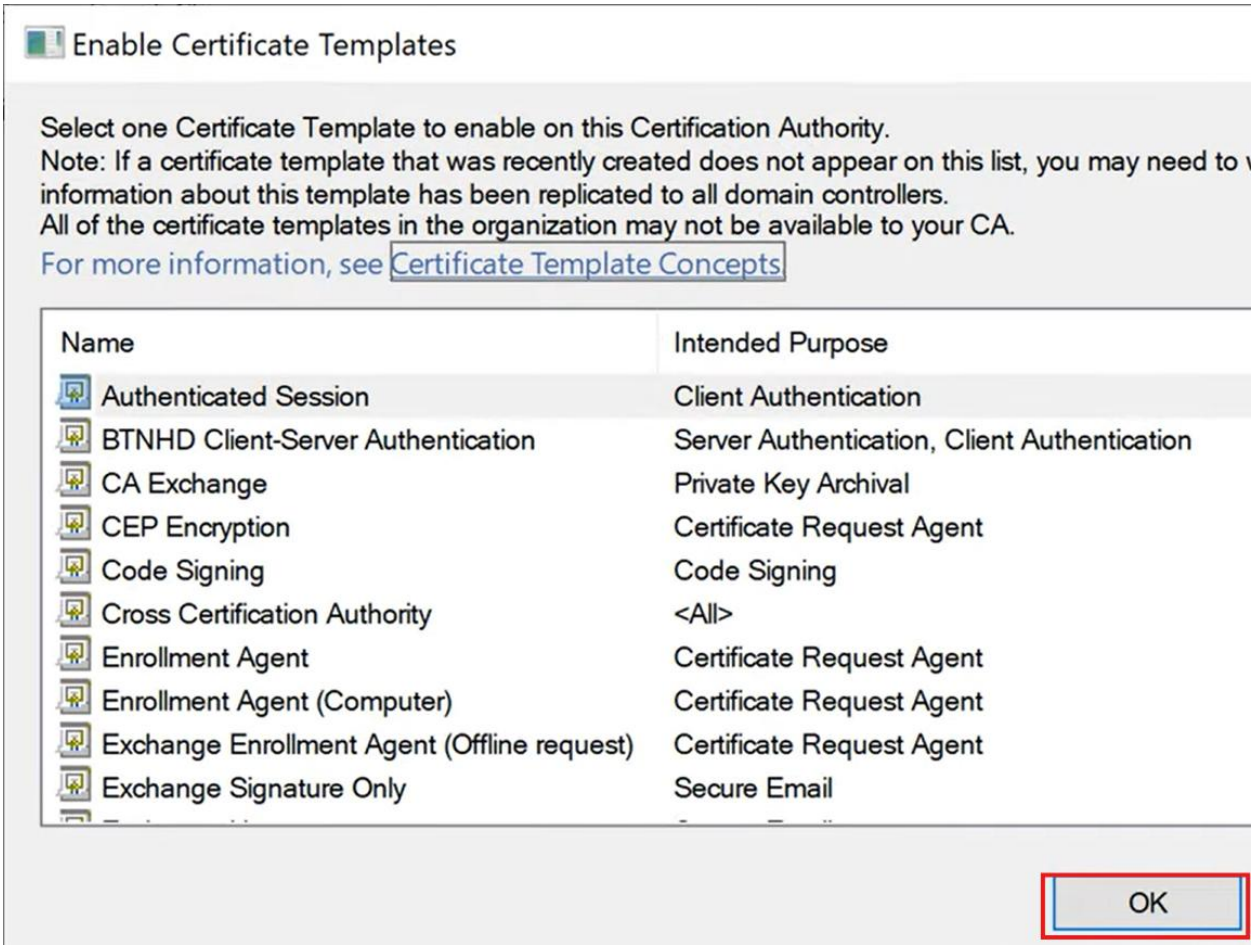
Note: Ensure Domain Controllers are included in Domain Computers group, otherwise DCs to receive the same permissions

Issuing the newly created Certificate Template

1. In **Certificate Authority (certsrv.msc)**, under our server, right-click **Certificate Templates**, and click **New > Certificate Template to Issue**



2. In the **Enable Certificate Templates**, select the newly created certificate template and click **Ok**



Verification

1. On any computer under the configured domain,
    a. Run **gpupdate /force**
    b. Run **certutil -pulse**
2. Open **Local Machine Certificate Store (certlm.msc)**
3. Check **Trusted Root Certification Authorities > Certificates** for **[INTERNAL-ORG]-ROOTCA-CA**
4. Check **Intermediate Certification Authorities > Certificates** for a **[INTERNAL-ORG]-INTERCA-CA** certificate
5. Check **Personal > Certificates** for a certificate issued by **[INTERNAL-ORG]-INTERCA-CA**.